Stock Market Prediction Using ARIMA and Machine Learning

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Overview

This paper explores the use of ARIMA (AutoRegressive Integrated Moving Average) and various Machine Learning (ML) techniques to predict stock market prices. Financial markets are complex, volatile systems influenced by many factors. The study reviews multiple forecasting approaches and concludes that combining ML with ARIMA improves prediction accuracy and supports more informed investment decisions.

Key Objectives

- Predict stock prices using ARIMA and machine learning models.
- Review existing research on AI-based stock market forecasting.
- Compare statistical and AI techniques in terms of predictive performance.
- Enable more accurate and informed investment decision-making.

Literature Review Highlights

Study	Technique	Key Finding
Poongodi et al. (2020)	Support Vector Machines (SVM)	SVM effectively predicts daily stock prices across markets of different sizes
Sudha et al. (2022)	Neural Networks, SVM, ARMA	All three AI methods outperformed linear models; random walk was the top
Hiransha et al. (2018)	Deep Learning / ANN	Deep learning models successfully forecast NSE (Indian) stock market pric
Dhyani et al. (2020)	ARIMA Hybrid Models	ARIMA outperforms traditional financial models (full history, single index, m

Methodology

The paper evaluates two primary categories of forecasting methods:

1. Statistical Techniques: ARIMA, ARCH, logistic regression, and ARMA models. ARIMA filters linear trends from data and passes residuals to ML models, enabling hybrid approaches that capture both linear and non-linear patterns.

2. Artificial Intelligence / Machine Learning Techniques: Including Multilayer Perceptrons (MLP), Convolutional Neural Networks (CNN), LSTM (Long Short-Term Memory), Support Vector Machines (SVM), Naive Bayes, and Recurrent Neural Networks (RNN).

LSTM networks use a three-gate architecture (Forget Gate, Input Gate, Output Gate) to selectively retain or discard information, making them well-suited for time series data. The Least Mean Squares (LMS) algorithm was also applied to demonstrate that linear methods can achieve reasonable accuracy in stock prediction tasks.

Datasets Used in Experiments

Dataset	Open (Min–Max)	Low (Min–Max)	High (Min–Max)	Close (Min–Max)
Google	87.74 – 1005.49	86.37 – 996.62	89.29 – 1008.61	87.58 – 1004.28
Nifty 50	7735.15 – 12932.5	7511.1 – 12819.35	8036.95 – 12948.85	7610.25 – 12938.25
Reliance	205.5 – 3298.0	197.15 – 3141.3	219.5 – 3298.0	203.2 – 3220.85

Results & Conclusion

The training phase takes approximately 10–15 minutes, while generating a prediction takes only a few seconds. Key outcomes of this research include:

- ARIMA hybrid models outperformed traditional financial models such as full history, constant correlation, single index, and multi-group models.
- All three AI techniques (NN, SVM, ARMA) outperformed linear models in forecasting financial index values.
- LMS demonstrated that linear algorithms can deliver accurate predictions despite the common preference for non-linear approaches.
- Machine learning integration with ARIMA simplifies and improves stock price prediction.
- The paper concludes that pricing analysis of the stock market will be significantly improved using combined ML and ARIMA approaches.

Key References

- Dhyani et al. (2020) — Stock market forecasting using ARIMA model. IJRTE, 8(6).
- Hiransha et al. (2018) — NSE stock market prediction using deep-learning models. Procedia Computer Science, 132.
- Moghar & Hamiche (2020) — Stock market prediction using LSTM RNN. Procedia Computer Science, 170.
- Poongodi et al. (2020) — Bitcoin price prediction using ARIMA model. IJITST, 10(4).
- Sudha et al. (2022) — Stock price prediction using machine learning. Journal of Algebraic Statistics, 13(3).